ZHIQI HUANG

SHENZHEN, CHINA / FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- Research Assistant at CUHK-Shenzhen and M.Phil. student at Waseda University, working toward task-relevant 3D simulation for sim-to-real embodied intelligence.
- My work follows one principle: preserve task-relevant structure while controlling the variation a system must handle. My background spans production graphics, controllable 3D generation, relightable materials, and persistence-aware volumetric forecasting.

RESEARCH INTERESTS

- Task-aware real-to-sim environments and controllable digital twins or digital cousins.
- Sim-real co-training, domain-gap diagnosis, adaptation, and transfer evaluation.
- Structured action-conditioned state models for contact-rich interaction.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen — Jul. 2026 - Present

Research Assistant — Shenzhen, China

- Research focus: task-relevant 3D simulation and sim-to-real embodied intelligence.

EDUCATION

Waseda University — Apr. 2025 - Mar. 2027 (expected)

M.Phil. in Information Architecture — Fukuoka, Japan

- English-taught program; GPA: 3.8 / 4.0

Sun Yat-sen University — 2017 - 2021

B.E. in Software Engineering — Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATION

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss — 2026

IEEE ICASSP 2026; First Author, Corresponding Author

- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Generates an editable 3D Gaussian head avatar from one image while preserving identity and enabling language-level control over expressions and accessories.
- Links: Project, IEEE Xplore, arXiv, Code

MANUSCRIPTS UNDER REVIEW

Controllable PBR material generation from long-form descriptions 2026

First Author; AAAI 2027 (Under Review)

- Generates relightable PBR channels by retaining detailed material specifications and routing them with surface geometry.
- Contributes to controllable simulation assets and systematic appearance variation.

Deform-then-edit forecasting for longitudinal 3D CT 2026

First Author; AAAI 2027 (Under Review)

- Forecasts localized volumetric change while explicitly preserving stable surrounding anatomy.
- Developed for medical imaging; provides experience in persistence-aware 3D transition modeling and strong no-change baselines.

INDUSTRY EXPERIENCE

4399 Games 2023 - 2024

Senior Graphics Engineer Guangzhou, China

- Led a 3-5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC.
- Developed reusable asset, material, and rendering pipelines for mobile, PC, and web projects.

4399 Games 2021 - 2023

Graphics Engineer Guangzhou, China

- Built and optimized the cross-platform real-time rendering pipeline for *Era of Conquest*, including shaders, PBR materials, and mobile performance profiling.

TECHNICAL SKILLS

- Graphics and systems: C++, C#, GLSL/HLSL; Unity, Vulkan, OpenGL, real-time rendering, PBR, cross-platform optimization.
- Machine learning and 3D: Python, PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh and material processing, multi-view evaluation, volumetric modeling.

LANGUAGES AND HONORS

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90)
- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University, 2018-2019